CS/S-25/CS4T003/02 . BT230020CS

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Education to Eternity

॥ज्ञानम् सर्वार्थ साधनम्॥

B. Tech Summer / ~~Winter~~ - 2025

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering **Sem:- IV**

Subject: FLAT **Subject Code: CS4T003**

Marks: 100 **Date of Exam: 02/05/2025 Time: 3 Hrs.**

Instructions to the students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

CO1: Define the mathematical principles behind theoretical computer science.

CO2: Differentiate and give examples for the different types of automata like finite automata, push down automata, linear bounded automata and turing machine.

CO3: Correlate the different types of automata to real world applications.

CO4: Design appropriate automata for the different requirements outlined by theoretical computer science.

CO5: Identify the different computational problems and their associated complexity.

SECTION A

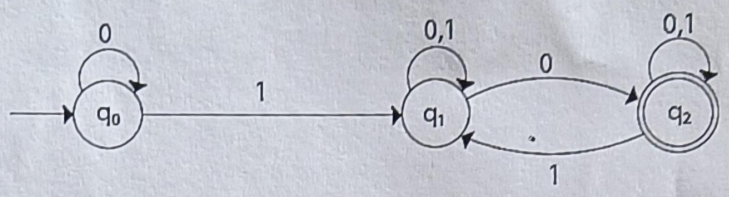
1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define the finite automata with neat diagram.	L1/CO1	2
Q.2	State various representations used in Transition diagram.	L1/CO1	2
Q.3	Give the definition of Regular set.	L1/CO1	2
Q.4	Differentiate positive closure and epsilon closure.	L1/CO1	2
Q.5	Write the pumping lemma theorem.	L3/CO2	2
Q.6	Identify the four classes of languages.	L2/CO2	2
Q.7	Distinguish between CNF and GNF.	L2/CO1	2
Q.8	Define Ambiguous Grammar.	L1/CO2	2
Q.9	Write a CFG to specify all strings over [a, b} that are palindromes.	L2/CO2	2
Q.10	Draw the PDA model.	L1/CO2	2
Q.11	Eliminate Ɛ-productions from the grammar: $S \rightarrow AaA$ $A \rightarrow Sb \mid bCC \mid \varepsilon$ $C \rightarrow CC \mid abb$	L3/CO2	2
Q.12	Define Turing machine.	L1/CO1	2

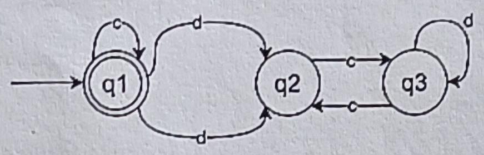
SECTION B

1. **All Questions are compulsory.**

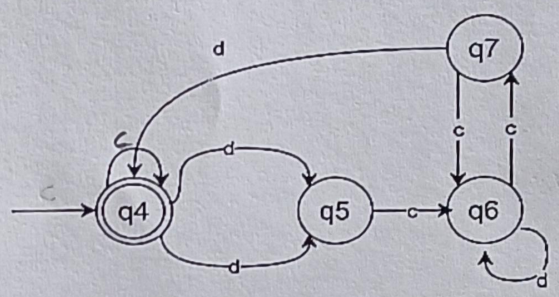
Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Convert the following NFA into its equivalent minimized DFA.	L3/CO3	4



Q.14	Construct the deterministic finite automata, which accepts the string starting with 1 and containing third position from the right side as 0. List the sample strings.	L2/CO4	4
Q.15	Find whether the following FSMs are equivalent or not.	L2/CO3	4



Machine A



Machine B

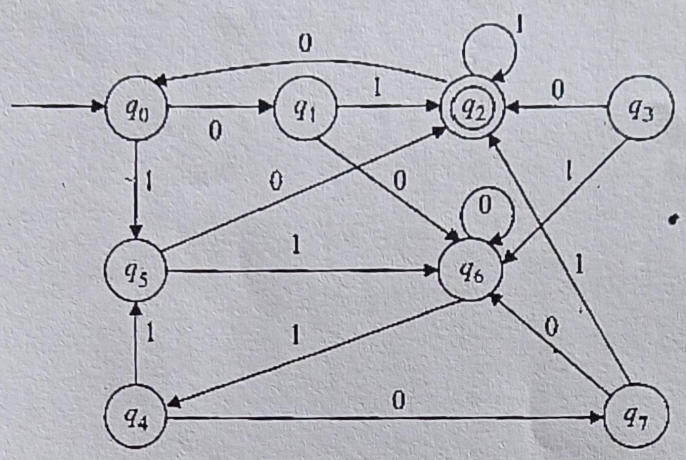
Q.16	Explain Multi tape Turing Machine with neat diagram.	L4/ CO2	4
Q.17	Define formally Type 0, Type 1, Type 2 and Type 3 grammar. Show the corresponding automata for each class.	L1/CO2	4
Q.18	Identify whether the following PCP has a solution or not?	L3/CO2	4

	A	B
i	w_i	x_i
1	10	101
2	011	11
3	101	011

SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Prove that following languages are not regular. **a)** $L= \{0^n1^n \mid n>=1\}$ **b)** $L= \{a^nb^{2n} \mid n>=0\}$	L5/CO4	7
Q.20	Convert the following grammar into Chomsky Normal Form. Check whether the grammar is in reduced form or not. $S \rightarrow abAB$ $A \rightarrow bAB \mid \lambda$ $B \rightarrow Baa \mid A \mid \lambda$	L3/CO3	7
Q.21	Minimize the following DFA.	L2/CO3	7



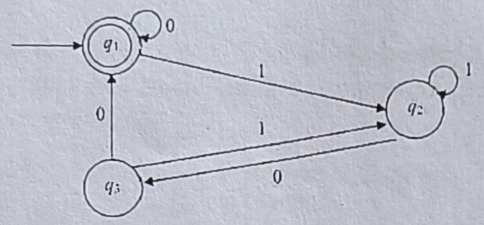
Q.22	Consider the following grammar. $S \rightarrow aSa \mid bSb \mid aAb \mid bAa$ $A \rightarrow Aa \mid Ab \mid \varepsilon$ Check whether the string 'abbbbaaba' accepted by the grammar. Find **a)** Rightmost Derivation **b)** Leftmost Derivation **c)** Parse Tree	L4/CO3	7
Q.23	Design a PDA for the language $L= \{anbmcn+m \mid (n, m)>=0\}$. Verify the sample string acceptance $a^n b^m c^{n+m}$	L4/CO4	7
Q.24	Design a Turing Machine for the language $\{ L = wcwR \mid n >=0\}$. Draw the machine also. wcw^R	L5/CO4	7

SECTION D

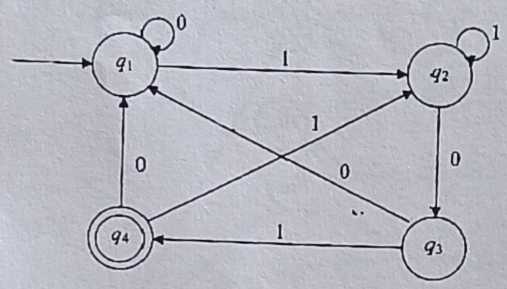
1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Design the following Moore machines:	L4/CO5	6
	a) Residue module 3 treated as a binary number		6
	b) Residue module 3 treated a decimal number		
Q.26	Find the regular expression for the following finite automata.	L3/CO2	12

a)



b)



Q.27	Construct a Context free grammar G which accepts following PDA	L3/CO5	12

for $M=(\{q_0, q_1\},\{a,b\},\{z_0,z\},\delta,q_0,z_0,\Phi)$

where δ is given by

$\delta(q_0,b,z0)=\{(q_0,zz)\}$

$\delta(q_0, \varepsilon,z0)=\{(q_0, \varepsilon)\}$

$\delta(q_0,b,z)=\{(q_0,zz)\}$

$\delta(q_0,a,z)=\{(q_1,z)\}$

$\delta(q_1,b,z)=\{(q_1, \varepsilon)\}$

$\delta(q_1,a,z_0)=\{(q_0,z_0)\}$